

CHIPPEWA VALLEY, WI, USA

WMO: 726435

Lat: 44.867N

Lon: 91.488W

Elev: 270

StdP: 98.13

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 14991

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.4	-22.5	-29.7	0.2	-24.7	-26.9	0.3	-21.9	9.4	-7.5	8.5	-7.2	2.1	250	0.566

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.7	32.1	22.8	30.4	21.7	28.8	20.7	24.4	29.6	23.3	28.3	22.3	27.1	5.2	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.7	18.0	27.5	21.7	16.9	26.3	20.6	15.8	25.2	75.0	29.6	70.7	28.3	66.6	27.1	29.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.1	8.1	7.3	-28.8	35.0	3.7	2.0	-31.5	36.4	-33.6	37.6	-35.7	38.8	-38.4	40.2		
			-29.0	26.2	3.6	1.3	-31.6	27.1	-33.7	27.9	-35.7	28.6	-38.4	29.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.3	-9.6	-7.3	-0.3	7.4	14.1	19.6	22.0	20.7	16.5	9.0	1.2	-6.2
	DBStd	12.28	7.23	6.95	6.79	5.52	4.67	3.78	3.20	3.10	4.63	4.92	5.71	6.62
	HDD10.0	2398	607	484	327	112	14	0	0	0	5	79	269	501
	HDD18.3	4339	865	718	578	329	147	29	6	12	89	291	515	759
	CDD10.0	1429	0	0	7	34	142	289	373	333	199	49	4	0
	CDD18.3	329	0	0	1	1	16	68	121	87	33	3	0	0
	CDH23.3	3127	0	0	3	27	207	658	1159	737	307	28	0	0
	CDH26.7	949	0	0	0	6	58	198	402	199	82	4	0	0
Wind	WSAvg	3.3	3.2	3.4	3.5	3.9	3.7	3.2	2.9	2.7	3.1	3.4	3.4	3.2
Precipitation	PrecAvg	780	25	18	42	77	105	109	89	111	92	56	38	27
	PrecMax	1154	88	45	107	154	233	190	176	296	223	143	137	66
	PrecMin	437	2	1	5	26	30	40	20	7	5	7	2	7
	PrecStd	171	22	12	23	38	46	38	45	65	54	36	27	18
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.8	10.9	21.6	26.9	31.5	33.6	35.3	33.1	31.8	26.6	19.1	10.9
		MCWB	3.5	7.3	14.2	16.5	20.0	23.2	24.3	23.3	22.2	18.3	12.6	8.3
	2%	DB	3.6	6.8	16.7	22.6	28.2	30.9	32.5	30.5	29.0	23.1	15.3	6.7
		MCWB	1.6	4.2	11.5	13.2	18.1	21.8	23.4	22.3	21.0	16.8	10.9	4.2
	5%	DB	1.7	4.1	13.0	19.9	25.6	28.9	30.5	28.8	26.9	20.2	12.4	4.0
		MCWB	0.5	2.2	8.2	11.2	16.7	20.2	22.3	21.2	19.6	14.6	8.4	2.2
	10%	DB	0.3	2.1	9.1	17.2	23.0	27.1	28.8	27.3	24.7	17.5	9.7	2.2
		MCWB	-0.7	0.5	5.7	9.8	15.1	19.4	21.2	20.3	18.5	13.1	6.6	0.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.9	7.2	15.8	17.3	22.5	24.8	26.7	25.4	23.7	21.0	14.3	8.8
		MCDB	6.1	10.4	19.9	24.0	29.3	31.1	32.2	30.4	28.9	23.9	17.0	10.4
	2%	WB	1.9	4.5	12.4	14.6	19.8	23.2	24.9	23.8	22.1	17.6	11.4	4.5
		MCDB	3.2	6.5	16.0	20.7	25.6	28.8	30.2	28.3	27.0	21.6	14.6	6.0
	5%	WB	0.7	2.3	8.8	12.5	18.0	21.9	23.6	22.8	20.8	15.5	8.8	2.3
		MCDB	1.6	4.0	12.5	17.9	23.2	27.0	28.8	27.0	25.2	19.1	11.9	3.8
	10%	WB	-0.7	0.5	5.7	10.5	16.5	20.6	22.4	21.6	19.5	13.3	6.7	0.9
		MCDB	0.3	2.0	8.8	16.5	21.6	25.4	27.4	25.7	23.7	17.0	9.6	2.0
Mean Daily Temperature Range	5% DB	MDBR	8.8	9.6	10.1	11.8	12.3	11.6	11.7	11.4	11.8	10.5	8.7	8.0
		MCDBR	9.0	10.5	14.2	16.8	15.8	13.7	13.2	12.7	13.8	14.8	13.2	10.1
		MCWBR	7.7	8.3	9.1	9.2	7.4	6.1	5.7	5.9	6.9	8.6	9.0	8.1
	5% WB	MCDBR	8.7	9.9	13.1	14.3	13.0	11.5	11.6	10.9	11.7	12.3	11.9	9.2
		MCWBR	7.6	8.0	8.8	8.9	7.0	6.1	5.8	6.0	6.5	8.2	8.9	7.5
Clear-Sky Solar Irradiance	taub		0.256	0.282	0.315	0.354	0.389	0.391	0.380	0.388	0.359	0.321	0.290	0.262
	taud		2.395	2.317	2.367	2.358	2.301	2.328	2.373	2.364	2.426	2.515	2.510	2.440
	Ebn at Noon		877	908	920	907	881	877	883	863	862	856	830	832
	Edh at Noon		72	95	105	116	127	124	117	114	99	79	66	63
All-Sky Solar Radiation	RadAvg		1.73	2.83	3.92	4.55	5.32	5.90	6.16	5.17	4.06	2.58	1.62	1.21
	RadStd		0.18	0.25	0.41	0.47	0.41	0.46	0.25	0.34	0.27	0.32	0.15	0.13

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (52 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page